

Vlasov-Poisson Equation with H :

$$\partial_t f + v \partial_x f - (E + H) \partial_v f = 0$$

where  $\partial_x E = 1 - \rho_f$ ,  $\rho_f(t, x) = \int_{\mathbb{R}} f(t, x, v) dv$ .

K-step:  $\partial_t K_i = \langle \text{RHS}_K(K(t, x), V^n), V_i^n \rangle_v$

$$= \langle -v \partial_x (K V^n) + (E + H) \partial_v (K V^n), V_i^n \rangle_v$$

$$= \langle -v \sum_{j=1}^J \partial_x K_j V_j^n + (E + H) \sum_{j=1}^J K_j \partial_v V_j^n, V_i^n \rangle_v$$

$$= - \sum_{j=1}^J \langle v V_j^n, V_i^n \rangle_v \partial_x K_j + (E + H) \sum_{j=1}^J \langle \partial_v V_j^n, V_i^n \rangle_v K_j$$

$$K^{n+1} = K^n + \Delta t \partial_t K^n$$

$$K^{n+1} \stackrel{\text{QR}}{=} U^{n+1} S^* T$$

S-step:  $\partial_t S_{ij}(t) = - \langle \text{RHS}_S(U^{n+1} S V^n), U_i^{n+1} V_j^n \rangle$

$$= - \langle \text{RHS} \left( \sum_{k=1}^J \sum_{l=1}^J U_k^{n+1} S_{kl}^* V_l^n \right), U_i^{n+1} V_j^n \rangle$$

$$= \sum_{k=1}^J \sum_{l=1}^J S_{kl} \langle \partial_x U_k^{n+1}, U_i^{n+1} \rangle_x \langle v V_l^n, V_j^n \rangle_v - \sum_{k=1}^J \sum_{l=1}^J S_{kl} \langle (E + H) U_k^{n+1}, U_i^{n+1} \rangle_x \langle \partial_v V_l^n, V_j^n \rangle_v.$$

$$S^{**} = S^* + \Delta t \partial_t S^n \Rightarrow L^n = V^n S^{**T}$$

L-step:  $\partial_t L_i^n = \langle \text{RHS}_L \left( \sum_k U_k^{n+1} L_k^n \right), U_i^{n+1} \rangle_x$

$$= -v \sum_{k=1}^J \langle \partial_x U_k^{n+1}, U_i^{n+1} \rangle_x L_k^n + \sum_{k=1}^J \langle E U_k^{n+1}, U_i^{n+1} \rangle_x \partial_v L_k^n$$

$$L^{n+1} = L^n + \Delta t \partial_t L^n$$

$$L^{n+1} \stackrel{\text{QR}}{=} V^{n+1} S^{n+1} T$$

set  $J(H) = \frac{1}{2} \| f(H)(T) - f^{\text{eq}} \|_{L^2(x,v)}^2$

$$J_r(H) = \frac{1}{2} ( \| f_r \|_{L^2}^2 - 2 \langle f_r, f^{\text{eq}} \rangle_{x,v} + \| f^{\text{eq}} \|_{L^2}^2 )$$

$$= \frac{1}{2} ( \| S \|_F^2 - 2 \langle S, \langle f^{\text{eq}}, (U_i(T) V_j(T))_{x,v} \rangle_{ij} \rangle_T + \| f^{\text{eq}} \|_{L^2}^2 )$$

$$J(f_r, H, P) = \frac{1}{2} \| f_r(T) - f^{\text{eq}} \|^2 + \int_0^T \langle g, F_r(f_r, H) - \partial_t f_r \rangle dt.$$

$$\delta_P J = 0 \Rightarrow \partial_t f_r = F_r(f_r, H) = P(-v \partial_x f_r + a(f_r, H) \partial_v f_r)$$

$$\delta_{f_r} J = 0 \Rightarrow \text{set } f_r^\epsilon = f_r + \epsilon \eta, \quad \eta|_{t=0} = 0.$$

$$\Rightarrow \delta J = \langle f_r(T) - f^{\text{eq}}, \eta(T) \rangle$$

$$\delta \int_0^T \langle g, F_r(f_r, H) - \partial_t f_r \rangle dt.$$

$$= \delta \int_0^T \langle g, D_{f_r} F_r[\eta] \rangle - \langle g, \partial_t \eta \rangle dt$$

$$= \delta \int_0^T \langle (D_{f_r} F_r)^* g, \eta \rangle + \langle \partial_t g, \eta \rangle dt - \langle g, \eta \rangle|_T.$$

thus,  $P(T) = f_r(T) - f^{\text{eq}}$

$$\partial_t g + (D_{f_r} F_r)^* g = 0$$

Discrete perspective

$$Z^n = (U^n, S^n, V^n), \quad Z^{n+1} = \Phi_n^{KSL}(Z^n, H^n)$$

$$J_n(Z^n) = \frac{1}{2} \| U^n S^n (V^n)^T - f^{\text{eq}} \|_h^2$$

In a forward step:

$$K_*^n = U^n S^n + h F_0^n W_v V^n$$

$$(U^{n+1}, \hat{S}^n) = \text{QR}_{W_{x,+}}(K_*^n)$$

$$\tilde{S}^n = \hat{S}^n - h (U^{n+1})^T W_x F_1^n W_v V^n$$

$$L_*^n = V^n (\tilde{S}^n)^T + h (F_2^n)^T W_x U^{n+1}$$

$$(V^{n+1}, (S^{n+1})^T) = \text{QR}_{W_{v,+}}(L_*^n)$$

$$\begin{aligned} \mathcal{L}_d = & \mathcal{J}_h(U^n, S^n, V^n) + \sum_{n=0}^{N-1} \langle P_K^n, U^n S^n + h F_0^n W_V V^n - K_K^n \rangle_F \quad (\mathcal{L}_K) \\ & + \langle P_S^n, \hat{S}^n - h(U^{nn})^T W_X F_1^n W_V V^n - \hat{S}^n \rangle_F \quad (\mathcal{L}_S) \\ & + \langle P_L^n, V^n (\hat{S}^n)^T + h(F_2^n)^T W_X U^{nn} - L_K^n \rangle_F \quad (\mathcal{L}_L) \end{aligned}$$

$$\delta \mathcal{L}_d = \delta \mathcal{J}_h + \sum_n (\delta \mathcal{L}_K^n + \delta \mathcal{L}_S^n + \delta \mathcal{L}_L^n)$$

$$\begin{aligned} \delta \mathcal{L}_K^n &= \langle P_K^n, \delta U^n S^n + U^n \delta S^n + h \delta F_0^n W_V V^n + h F_0^n W_V \delta V^n - \delta K_K^n \rangle_F + \langle \delta P_K^n, \dots \rangle \\ \delta \mathcal{L}_S^n &= \langle P_S^n, \delta \hat{S}^n - h \delta(U^{nn})^T W_X F_1^n W_V V^n - h(U^{nn})^T W_X \delta F_1^n W_V V^n \\ &\quad - h(U^{nn})^T W_X F_1^n W_V \delta V^n - \delta \hat{S}^n \rangle_F + \langle \delta P_S^n, \dots \rangle \\ \delta \mathcal{L}_L^n &= \langle P_L^n, \delta V^n (\hat{S}^n)^T + h \delta(F_2^n)^T W_X U^{nn} + h(F_2^n)^T W_X \delta U^{nn} - \delta L_K^n \rangle_F + \langle \delta P_L^n, \dots \rangle \\ \delta \mathcal{J}_h &= \langle w \delta U^n (S^n)^T, \delta U^n \rangle_F + \langle w(U^n)^T \delta V^n, \delta S^n \rangle_F + \langle w \delta U^n S^n, \delta V^n \rangle_F \\ &\quad (G = U^n S^n (V^n)^T - f^{eq} \quad w = \Delta x \Delta v) \end{aligned}$$

$\delta L_K$

$L_K$  related to  $\delta L_K^n, \delta V^{nn}, \delta S^{nn+1}$

$$\begin{aligned} \delta L_K^n \mathcal{L}_d &= -\langle P_L^n, \delta L_K^n \rangle + \delta L_K^n \mathcal{L}_K^{nn+1} + \delta L_K^n \mathcal{L}_S^{nn+1} + \delta L_K^n \mathcal{L}_L^{nn+1} \\ \delta L_K^n \mathcal{L}_K^{nn+1} &= \langle U^{nn+1} P_K^{nn+1}, \delta S^{nn+1} \rangle + \langle h W_V (F_0^{nn+1})^T P_K^{nn+1}, \delta V^{nn+1} \rangle + \langle h \hat{P}_0^{nn+1} (V^{nn+1})^T W_V, \delta F_0^{nn+1} \rangle \\ &= \langle U^{nn+1} P_K^{nn+1} + U^{nn+1} \hat{P}_0^{nn+1} V^{nn+1}, \delta S^{nn+1} \rangle + \langle h W_V (F_0^{nn+1})^T P_K^{nn+1} + \hat{P}_0^{nn+1} U^{nn+1} S^{nn+1}, \delta V^{nn+1} \rangle \\ &= \langle \hat{P}_0^{nn+1}, \delta \hat{P}_0^{nn+1} \rangle \\ &= \langle \hat{P}_0^{nn+1}, \delta(U^{nn+1} S^{nn+1} V^{nn+1}) \rangle \\ \delta L_K^n \mathcal{L}_S^{nn+1} &= \langle -h W_V (F_1^{nn+1})^T W_X U^{nn+2} P_S^{nn+1}, \delta V^{nn+1} \rangle + \langle -h W_X U^{nn+2} P_S^{nn+1} V^{nn+1} W_V, \delta F_1^{nn+1} \rangle \\ &= \langle -h W_V (F_1^{nn+1})^T W_X U^{nn+2} P_S^{nn+1} + (g_1^{nn+1})^T U^{nn+2} \hat{S}^{nn+1}, \delta V^{nn+1} \rangle \\ \delta L_K^n \mathcal{L}_L^{nn+1} &= \langle P_L^{nn+1} \hat{S}^{nn+1}, \delta V^{nn+1} \rangle + \langle h W_X U^{nn+2} (P_L^{nn+1})^T, \delta F_2^{nn+1} \rangle \\ &= \langle P_L^{nn+1} \hat{S}^{nn+1} + (g_2^{nn+1})^T U^{nn+2} \hat{S}^{nn+1}, \delta V^{nn+1} \rangle \end{aligned}$$

$$\text{then we get } \delta L_K^n \mathcal{L}_d = -\langle P_L^n, \delta L_K^n \rangle + \langle \bar{S}^{nn+1}, \delta S^{nn+1} \rangle + \langle \bar{V}^{nn+1}, \delta V^{nn+1} \rangle$$

$$\langle U^{nn+1}, (S^{nn+1})^T \rangle = \text{DR}_{W_V, +} (L_K^{n+1}), \text{ set } (\delta V^{nn+1}, \delta(S^{nn+1})^T) = \text{DR}_{W_V, +} (L_K^{n+1}) \mathcal{L}_K^{nn+1}$$

$$\text{we define that } \delta V = \text{DR}(L_K) \mathcal{L}_K, \delta S = (\text{DR}(L_K) \mathcal{L}_K)^T$$

$$\text{thus, } P_L^n = \bar{L}_K^n = \text{DR}(L_K^n)^* \bar{V}^{nn+1} + \text{DR}(L_K^n)^* (\bar{S}^{nn+1})^T$$

$$(\text{DR}^* \& \text{DR}^* \text{ is the inverse of DR, details in next pages})$$

$\delta \tilde{S}$ :

$$\delta \tilde{S} \mathcal{L}_S = -\langle P_S, \delta \tilde{S} \rangle$$

$$\delta \tilde{S} \mathcal{L}_L = \langle P_L, V \delta \tilde{S}^T \rangle + h \langle P_L, \delta F_2^T W_X U_1 \rangle$$

$$\delta F_2 = \text{DR}_{\text{RHS}}(f_2, H) \mathcal{L}_U \delta \tilde{S} V^T$$

$$\Rightarrow h \langle P_L, \delta F_2^T W_X U_1 \rangle_F = \langle h W_X U_1 P_L^T, \delta F_2 \rangle$$

$$= \langle g_2, \delta f_2 \rangle$$

$$= \langle U_1^T g_2 V, \delta \tilde{S} \rangle$$

$$(U_1 \text{ refers to } U_{n+1})$$

$$\Rightarrow \delta \tilde{S} \mathcal{L}_d = -\langle P_S + P_L^T V + U_1^T g_2 V, \delta \tilde{S} \rangle_F = 0$$

$$\text{i.e. } P_S = P_L^T V + U_1^T g_2 V$$

$\delta K_K$ :

$$K_K, U^{nn}, \hat{S}^n$$

$$\delta K_K^n \mathcal{L}_S = \langle P_S^n, \delta \hat{S}^n \rangle_F$$

$$-h \langle P_S^n, (S U^{nn})^T W_X F_1^n W_V V^n \rangle_F = \langle -h W_X F_1^n W_V V^n (P_S^n)^T, \delta U^{nn} \rangle_F$$

$$-h \langle P_S^n, (U^{nn})^T W_X \delta F_1^n W_V V^n \rangle_F = \langle g_1^n, \delta f_1^n \rangle_F = \langle g_1^n V^n (\hat{S}^n)^T, \delta U^{nn} \rangle + \langle (U^{nn})^T g_1^n V^n, \delta \hat{S}^n \rangle_F$$

$$\delta K_K^n \mathcal{L}_L = h \langle P_L^n, (\delta F_2^n)^T W_X U^{nn} \rangle_F + h \langle P_L^n, (F_2^n)^T W_X \delta U^{nn} \rangle_F$$

$$= \langle g_2^n, \delta f_2^n \rangle_F + \dots$$

$$= \langle g_2^n, \delta U^{nn} \hat{S}^n (V^n)^T \rangle_F + \langle h W_X F_2^n P_L^n, \delta U^{nn} \rangle_F$$

$$= \langle h W_X F_2^n P_L^n + g_2^n V^n (\hat{S}^n)^T, \delta U^{nn} \rangle$$

$$f_0^n = U^n S^n (V^n)^T \quad F_0^n = \text{RHS}(f_0^n, H)$$

$$f_1^n = U^{nn} \hat{S}^n (V^n)^T$$

$$f_2^n = U^{nn} \hat{S}^n (V^n)^T$$

$$\delta f_0 = \delta U S V^T + U \delta S V^T + U S \cdot \delta V^T$$

$$\delta F_0 = \text{DR}_{\text{RHS}}(f_0, H) \mathcal{L}_U \delta f_0$$

$$(S U^{nn}, \delta(S^{nn})^T) = \text{DR}_{W_V, +} (L_K^n) \mathcal{L}_K^n$$

$$(S V^{nn}, \delta(S^{nn})^T) = \text{DR}_{W_V, +} (L_K^n) \mathcal{L}_K^n$$

$$\bar{F}_0 = h P_K V^T W_V$$

$$\langle P_K, h \delta F_0 W_V V \rangle_F$$

$$= h \text{tr}(P_K^T \delta F_0 W_V V)$$

$$= \langle h P_K V^T W_V, \delta F_0 \rangle_F$$

$$\Rightarrow \bar{F}_0 = h P_K V^T W_V$$

$$\bar{F}_1 = -h W_X U_1 P_S^T W_V \quad \bar{F}_2 = h W_X U_1 P_L^T$$

$$g_a = \bar{f}_a := (\text{DR}_{\text{RHS}}(f_a, H))^* \bar{F}_a$$

$$\langle \bar{F}_a, \delta F_a \rangle = \langle \bar{f}_a, \delta f_a \rangle_{\delta(U S V^T)}$$

$$(\text{Coefficient of } \delta U^{n+1}) \quad C_U = \bar{U}^{n+1} - h W_x F_1^n W_x V^n (P_S^n)^T + g_1^n V^n (\hat{S}^n)^T + h W_x F_2^n P_L^n + g_2^n V^n (\hat{S}^n)^T$$

$$\begin{cases} \bar{U}^N = \Delta x \Delta v \sum U^N S^N (V^N)^T - f^{eq} \int V^N (S^N)^T \\ \bar{U}^{n+1} = P_K^{n+1} (S^{n+1})^T + g_0^{n+1} V^{n+1} (S^{n+1})^T \end{cases}$$

$$(\text{Coefficient of } \delta \hat{S}) \quad C_{\hat{S}} = P_S^n + (U^{n+1})^T g_1^n V^n$$

$$\begin{aligned} \Rightarrow \delta K_* I_d &= - \langle P_K, \delta K_* \rangle_F + \langle C_U, \delta U^{n+1} \rangle_F + \langle C_{\hat{S}}, \delta \hat{S} \rangle_F = 0 \\ &\quad \langle C_U, DR [\delta K_*] \rangle_F + \langle C_{\hat{S}}, DR [\delta K_*] \rangle_F \\ \Rightarrow P_K &= (DR)^* C_U + (DR)^* C_{\hat{S}} \end{aligned}$$

$$\delta_H : \quad \delta f_a^n = 0 \quad a=0,1,2$$

$$\delta_H F_a^n = D_{HRS} (f_a^n, H) \int \delta_H J$$

$$\delta_H I_a = \sum_{n=0}^{N-1} \sum_{a=0}^2 \langle \bar{F}_a^n, D_{HRS} (f_a^n, H) \int \delta_H J \rangle_F$$

$$= \langle \sum_{n=0}^{N-1} \sum_{a=0}^2 (D_{HRS} (f_a^n, H))^* \bar{F}_a^n, \delta_H J \rangle$$

$$\Rightarrow \nabla_H J_H = \sum_{n=0}^{N-1} \sum_{a=0}^2 (D_{HRS} (f_a^n, H))^* \bar{F}_a^n$$

Initial condition

$$J = \frac{\Delta x \Delta v}{2} \| U^N S^N (V^N)^T - f^{eq} \| \quad \text{Def } a = U^N S^N (V^N)^T - f^{eq}$$

$$\delta J = \frac{\langle w a V^n (S^n)^T, \delta U^n \rangle}{\bar{U}^N} + \frac{\langle w (U^n)^T a V^n, \delta S^n \rangle}{\bar{S}^N} + \frac{\langle w a^T U^n S^n, \delta V^n \rangle}{\bar{V}^N}$$

$$\text{Then we get } P_L^{N-1} = DR (L_*^{N-1})^* [\bar{V}^N] + DR (L_*^{N-1})^* [\bar{S}^N]^T$$

Inverse QR

$$A = QR \quad Q^T W Q = I$$

$$\begin{cases} \delta A = \delta QR + Q \delta R \\ ( \delta Q )^T W Q + Q^T W \delta Q = 0 \end{cases}$$

$$( \delta Q )^T W Q + Q^T W \delta Q = 0$$

$$\Rightarrow Q^T (W I) \delta A R^{-1} = Q^T (W I) \delta Q + \delta R R^{-1}$$

$$I + I : \quad \delta R R^{-1} + (\delta R R^{-1})^T = Q^T (W I) \delta A R^{-1} + (Q^T (W I) \delta A R^{-1})^T$$

$$\text{If } \delta R R^{-1} \text{ is skew} \Rightarrow (\delta R R^{-1})_{ij} = \begin{cases} 0 & i=j \\ (Q^T (W I) \delta A R^{-1})_{ii} & i=j \\ (Q^T (W I) \delta A R^{-1})_{ij} & i < j \\ -(Q^T (W I) \delta A R^{-1})_{ji} & i > j \end{cases}$$

$$\Rightarrow \delta R = U (Q^T W \delta A R^{-1}) R \quad \text{i.e. } \delta R = DR(A) \int \delta A J$$

$$\Rightarrow \delta Q = \delta A R^{-1} - Q U (Q^T W \delta A R^{-1})$$

$$[DR]^* : \quad \text{Def } S(N) := \text{copy}(N) \quad S(N)_{ij} = \begin{cases} N_{ij} & i=j \\ N_{ji} & i < j \end{cases}$$

$$\langle N, U(M) \rangle = \langle S(N)^T, M \rangle$$

$$\langle B, \delta Q \rangle = \langle B, \delta A R^{-1} \rangle - \langle Q^T B, U(Q^T W \delta A R^{-1}) \rangle$$

$$= \langle [B - WQS(B^T Q)] R^{-1}, \delta A \rangle$$

$$\langle C, \delta R \rangle = \langle C R^T, U(Q^T W \delta A R^{-1}) \rangle$$

$$= \langle WQS(RC^T) R^{-1}, \delta A \rangle$$

$$g_a = \bar{f}_a$$

$$\langle B, \delta f \rangle = \langle B, -v D_x \delta f + E D_v \delta f + \delta E D_v f \rangle \quad (D_x^* = -D_x \quad D_v^* = -D_v \quad \text{in central discretization})$$

$$\langle B, \delta E D_v f \rangle = \sum_i \delta E_i \sum_j B_{ij} (D_v f)_{ij} \bar{E}$$

$$(\delta p_i = -\Delta v \sum_j \delta f_{ij}, \delta p_i = -\Delta v \sum_j \delta f_{ji}, \delta E = P \delta p) \quad \langle \bar{E}, \delta E \rangle = \langle P^* \bar{E}, \delta p \rangle = \langle \bar{p}, \delta p \rangle = \sum_{ij} [-\Delta v (\bar{p}_i)_j] \delta f_{ij}$$

$$\Rightarrow g_{ij}^{(E)} = \bar{f}_{ij}^{(E)} = -\Delta v (\bar{p})_i$$